Unifying Theory of Dynamic Aether Mechanics: Gravity

Erik Otto

Abstract

Within the Dynamic Aether Mechanics framework, gravity arises as the residual asymmetry of a transient binding cycle in which virtual particle–antiparticle pairs form and annihilate continuously at every atom. This paper develops the gravitational mechanism, derives the gravitational constant G from binding physics, and resolves the vacuum catastrophe. The framework is extended to gravitomagnetism (frame dragging, confirmed by Gravity Probe B), gravitational lensing (deriving the full general-relativistic bending angle $4GM/(r_0c^2)$ from two independent mechanisms), and Cherenkov radiation. Connections to mass–energy equivalence and the self-limiting behavior of gravity are established.

This paper is part of a series presenting the Unifying Theory of Dynamic Aether Mechanics. For the foundational concepts of Aether binding, the three independent properties (density, pressure, temperature), and the unification of force constants, see Paper 1: General Overview [12].

1. Gravity

Gravity in this theory is a true wave—an oscillating pressure disturbance propagating through Aether, generated by the transient bind–unbind cycling that occurs continuously at every atom. The mechanism corresponds to the virtual particle–antiparticle fluctuations described by quantum field theory, reinterpreted as the physical process of Aether transiently binding and unbinding.

1.1 The Mechanism

The spinning vortex core of each electron acts as a rapidly moving boundary in the sense of the Dynamic Casimir Effect [1]: the circulation velocity near the healing length ($\xi \approx 0.2$ fm) is sufficient to enhance vacuum fluctuations into transient particle–antiparticle pairs—virtual pairs in the language of quantum field theory. This is the same DCE mechanism that drives permanent binding (Section 1.5), operating below the permanence threshold. Each binding event consumes Aether from the local region, creating a brief low-pressure pulse. The pair is unstable and almost immediately annihilates, restoring the consumed Aether and returning the local pressure to normal. The vortex circulation then drives the next cycle.

Each cycle produces one pulse of the gravitational wave: a dip to low pressure followed by a return to normal. At the extraordinarily high cycling frequencies implied by atomic-scale processes, this oscillation would be indistinguishable from a static attractive force at any macroscopic measurement scale—consistent with our observation that gravity appears to be a constant, static field.

The wave propagates outward at the speed of light, since light speed represents the propagation speed of compression waves in Aether. This is consistent with the observed speed of gravitational waves.

1.2 Why Gravity Is Attractive

The pressure wave produced by transient cycling is asymmetric: it oscillates between normal pressure and low pressure, never exceeding normal pressure. The binding event consumes Aether (low pressure), and the annihilation merely restores it (normal pressure). The time-averaged effect is therefore a net reduction in pressure near the source—a persistent inward pull on surrounding matter.

1.3 The Gravitational Constant and the Uncertainty Principle

Each atom cycles independently, so the total gravitational wave output of a mass is proportional to the number of atoms it contains—proportional to its mass M. The gravitational constant G, in this framework, encodes properties of the transient cycling process:

- The mass of the transiently formed particle pair (how much Aether is consumed per cycle)
- The cycling frequency (how many bind–unbind events occur per second)
- The minimum vortex velocity at which transient binding is triggered
- The ratio of transient to permanent binding events (governed by the DCE permanence threshold)

The total force on a unit of mass in m_2 from all units in m_1 is $m_1 G$. Summed across all units of m_2, the force at zero radius becomes Gm_1m_2. Since the pressure wave dissipates radially, the force at distance r becomes Gm_1m_2/r^2—exactly as observed.

The cycling frequency and transient particle mass are not arbitrary. In quantum field theory, virtual particle formation is governed by the Heisenberg uncertainty principle [5]:

$$\Delta E \cdot \Delta t \geq \frac{\hbar}{2} \tag{1}$$

where ΔE is the energy of the virtual pair (related to its mass by $E = mc^2$) and Δt is the duration of its existence. More massive virtual pairs exist for shorter durations. Less massive pairs can persist longer. This relationship constrains the cycling parameters:

- If the transient particle mass is large, it exists very briefly and cycles rapidly.
- If the transient particle mass is small, it exists longer and cycles more slowly.

In either case, the product of mass and cycling frequency is constrained by $\hbar$. The Dynamic Casimir Effect further constrains the system by defining the vortex velocity at which transient events become permanent. Together, these constraints mean G may ultimately be derivable from $\hbar$, c, and the specific properties of Aether (density, vortex velocity threshold)—connecting gravity to quantum mechanics through fundamental constants. The difficulty of reconciling gravity with quantum mechanics (the long-sought theory of quantum gravity) may stem from not recognizing that they share a common mechanism: the transient formation and annihilation of particles from a dynamic medium, governed by the same uncertainty principle and DCE threshold conditions.

G may not be truly constant. Since $G \propto \epsilon\, \rho_A\, \psi^2$ (as derived in the Unification of Force Constants section), it could in principle vary through two channels: the local Aether density ρ_A and the asymmetry fraction ϵ. However, the Bernoulli analysis shows that ρ_A is essentially constant near gravitational sources ($\Delta\rho_A/\rho_A = 0$ for free-fall inflow), and the cosmological density variation is only $\sim 10^{-4}$—negligible. The primary channel for G variation is therefore through ϵ, which depends on temperature: higher temperature provides thermal energy that resists binding (analogous to how heating prevents condensation), potentially reducing the fraction of each cycle's energy that escapes as the gravitational wave. Cold environments (near matter, where thermal energy is locked into mass formation) could have higher ϵ and therefore stronger gravitational output, while hot environments (cosmic voids) could have lower ϵ.

1.4 Self-Limiting Behavior

A potential concern is runaway feedback: if a transiently formed particle pair has mass, and mass generates gravity, could transient particles trigger additional transient binding in a cascading loop?

The Heisenberg uncertainty principle provides the primary self-limiting mechanism. A virtual pair of energy ΔE can only exist for a time $\Delta t \leq \hbar/(2\Delta E)$. The more massive the transient pair, the shorter its existence. A transient particle cannot persist long enough to complete its own gravitational cycling—its lifetime is constrained by the uncertainty principle to be far shorter than the period of any gravitational wave it could produce. The cascade is cut off by the same quantum mechanical constraint that governs the cycling in the first place.

The Dynamic Casimir Effect reinforces this: for a transient particle to trigger further permanent binding, it would need to act as a "moving boundary" with sufficient velocity to exceed the DCE threshold. But a transient particle exists for too brief a duration and has no persistent spin vortex—it lacks the sustained, directional motion required to function as a DCE boundary. Only permanent particles with stable electron spin can sustain the vortex dynamics needed for continuous cycling.

Additional self-limiting factors include:

- The annihilation of the transient pair produces a restoring pressure pulse that disrupts the circulation pattern needed to sustain further binding nearby.
- The DCE threshold acts as a natural regulator: only Aether circulated by a persistent vortex (a permanent electron) undergoes cycling. A fleeting fluctuation from a transient pair does not sustain the boundary motion long enough to trigger secondary binding.

1.5 Connection to $E = mc^2$

The transient binding mechanism connects directly to mass–energy equivalence. Each bind–unbind cycle momentarily converts energy into bound mass and back. The energy recycled per cycle is mc^2 of the transiently formed particle pair (as derived in the Special Relativity section below). In QFT terms, the energy is "borrowed" from the vacuum for duration Δt and returned upon annihilation.

The gravitational wave's energy is a tiny fraction of mc^2—only the net outward-propagating pressure disturbance, not the full binding energy. This is why gravity is so extraordinarily weak compared to other forces: the vast majority of each cycle's energy is recycled locally, and only a minuscule asymmetry (the difference between the low-pressure binding pulse and the normal-pressure restoration) propagates as the gravitational wave.

When the DCE threshold is exceeded and a binding event becomes permanent, the full mc^2 of energy is locked into the resulting particle rather than being returned. This is the energy cost of mass formation, paid by the local environment (the vortex kinetic energy and surrounding Aether thermal energy).

1.6 The Vacuum Catastrophe

The identification of transient binding with virtual particle fluctuations offers a perspective on one of the largest unsolved problems in physics: the vacuum catastrophe.

When physicists use quantum field theory to calculate the energy density of vacuum fluctuations, they obtain a value roughly 10^{120} times larger than the observed cosmological constant (the measured energy density of empty space). This is often called the worst prediction in the history of physics.

This theory suggests a resolution. The enormous energy of vacuum fluctuations does exist—it is the total energy being cycled through transient binding events at every point in space. But nearly all of this energy is recycled locally. Each binding event invests mc^2 and recovers mc^2 upon annihilation. The net energy that propagates outward as the gravitational wave is only the tiny asymmetry of the cycle—the difference between the low-pressure dip and the normal-pressure restoration.

The vacuum catastrophe arises from counting the total energy of fluctuations. The physically observable effect (gravity, and by extension the cosmological constant) comes only from the residual asymmetry. The ratio between the two—the fraction of vacuum energy that actually propagates rather than being recycled—could account for the enormous discrepancy. The vacuum energy is real, but its gravitational effect is not its total magnitude; it is only the unrecycled remainder.

1.7 Gravitomagnetism: Frame Dragging from Circulatory Aether Flow

The v^2/c^2 universality established in Section 3 has a direct gravitational consequence. In electromagnetism, a moving charge sweeps its flow vector (the electric field) into angular structure (the magnetic field), with $F_{\mathrm{mag}} = F_{\mathrm{elec}} \times v_1 v_2/c^2$.[1] The same mechanism applies to any source of Aether perturbation that is set into motion—including gravity.

A mass element moving at velocity $\mathbf{v}$ drags its locally bound Aether, creating angular flow structure that stands in the same relation to the gravitoelectric flow vector $\mathbf{g}$ as the magnetic flow angle $\mathbf{B}$ stands to the electric flow vector $\mathbf{E}$:

$$\mathbf{B}_g \;=\; -\frac{1}{c^2}\,\mathbf{v}\times\mathbf{g}\,. \tag{2}$$

This is the gravitomagnetic field. No new parameters are introduced; the coupling is fixed entirely by the ratio $v^2\rho_A/K_A = v^2/c^2$.

For a body with angular momentum $\mathbf{J}$, integration over the mass–current distribution yields a dipole gravitomagnetic field identical in structure to the magnetic dipole field of a current loop:

$$\mathbf{B}_g(\mathbf{r}) \;=\; \frac{G}{c^2\,r^3}\left[3(\mathbf{J}\cdot\hat{\mathbf{r}})\,\hat{\mathbf{r}} - \mathbf{J}\right]. \tag{3}$$

A gyroscope with spin angular momentum $\mathbf{S}$ placed in this field precesses at the Larmor rate

$$\boldsymbol{\Omega}_{\mathrm{FD}} \;=\; \frac{1}{2}\,\mathbf{B}_g\,, \tag{4}$$

where the factor of $\frac{1}{2}$ is the standard gyroscopic coupling (spin precesses at half the imposed rotation rate).

[1] Paper 4 [14] identifies the electromagnetic "flow vector" concretely as the coherent axial-oscillation wave field of the bound core ring; the gravitational analog here uses the same flow-vector/flow-angle dichotomy at the macroscopic level.

Quantitative prediction for Gravity Probe B. The Gravity Probe B (GP-B) satellite placed four cryogenic gyroscopes in a polar orbit at altitude $h = 642$ km (orbital radius $R = 7.013 \times 10^6$ m) above the spinning Earth ($J_\oplus = I_\oplus \omega_\oplus = 5.86 \times 10^{33}$ kg m^2 s^{-1}). The gyroscope spin axes were aligned with the guide star IM Pegasi (declination $\delta = 16.84°$). Averaging over the polar orbit, the frame-dragging precession in the west–east direction is

$$\Omega_{\rm FD} = \frac{G J_\oplus}{2 c^2 R^3} \cos\delta = 39.3 \text{ mas yr}^{-1}, \tag{5}$$

where 1 mas $= 4.85 \times 10^{-9}$ rad. This is a *no-fit derivation*: it follows entirely from the v^2/c^2 universality with zero adjustable parameters.

The GP-B experiment measured $\Omega_{\rm FD} = -37.2 \pm 7.2$ mas yr^{-1}, in agreement with the prediction within 0.3σ. General relativity predicts -39.2 mas yr^{-1} from the weak-field limit of the Kerr metric—the same formula, confirming that the Aether mechanism reproduces the Lense–Thirring effect exactly. The LAGEOS satellite tracking experiments independently confirmed frame dragging to approximately 10% precision via orbital node precession.

Physical interpretation. The Aether theory provides a concrete mechanical picture for frame dragging. A rotating mass continuously drags its permanently bound Aether into co-rotation. This angular flow structure is structurally identical to the pattern that produces magnetism from moving charges—in both cases, motion converts a flow vector into a flow angle—differing only in the coupling mechanism (mass binding versus charge flow) and therefore in overall strength (suppressed by $\varepsilon \sim 10^{-43}$ relative to EM).

This unifies the concept of "frame dragging" discussed in Section 2 (Black Holes) with the magnetic force mechanism of Section 1 (Magnetism): both are circulatory Aether flow from a moving source, governed by the universal v^2/c^2 ratio. For black holes, the same mechanism drives relativistic jet collimation, where the central rapidly rotating mass creates gravitomagnetic fields strong enough to redirect infalling matter along the polar axis.

Fourth v^2/c^2 phenomenon. Gravitomagnetism extends the v^2/c^2 universality from three confirmed phenomena to four:

(a) **Time dilation**: $\gamma = 1/\sqrt{1 - v^2/c^2}$ — geometric path-length effect,

(b) **Magnetic force**: $F_{\rm mag} = F_{\rm elec} \times v_1 v_2/c^2$ — flow angle from charge motion,

(c) **Relativistic energy**: $E = \gamma m c^2$ — compression energy,

(d) **Gravitomagnetism**: $B_g = g \times v/c^2$ — flow angle from mass motion; confirmed by GP-B.

All four are manifestations of the single dimensionless ratio $v^2\rho_A/K_A = v^2/c^2$: kinetic energy density divided by the elastic capacity of the Aether medium.

2. Local Effects: Cherenkov Radiation and Gravitational Lensing

These phenomena are explained by the Aether inflow environment near matter.

Cherenkov radiation is a flash of light that occurs when a charged particle traveling at near light speed enters a material medium with a lower speed of light, such as water or glass. In such materials, the bound atomic structure modifies the Aether's wave propagation properties: the effective wave speed is reduced to c/n, where n is the material's refractive index. When a fast particle enters this region and exceeds the local wave propagation speed, it produces a compression shockwave—a "lucid boom"—that either is the observed light itself or causes the release of photons by disturbing nearby electrons. Note that this is a property of *material* media (bound matter), not of the gravitational inflow: the free Aether maintains $c = \sqrt{K_A/\rho_A}$ throughout the gravitational field, as shown by the Bernoulli analysis below.

2.1 Gravitational Lensing

Gravitational lensing is explained by the acoustic metric of the Aether inflow.

2.1.1 The Acoustic Metric

Light propagating through a flowing medium follows geodesics of the *acoustic metric*—the effective spacetime geometry experienced by waves in a moving medium. For the gravitational inflow at $v_{\text{in}}(r) = \sqrt{2GM/r}$ with uniform wave speed c, the acoustic metric takes the Painlevé–Gullstrand form:

$$ds^2 = \left(c^2 - v_{\text{in}}^2\right) dt^2 - 2\, v_{\text{in}}\, dr\, dt - dr^2 - r^2\, d\Omega^2 \tag{6}$$

This is mathematically equivalent to the Schwarzschild metric of general relativity, expressed in "river" coordinates where the Aether flows inward at escape velocity. All predictions of the Schwarzschild geometry—including gravitational lensing, time dilation, and the event horizon—follow directly from this acoustic metric without invoking spacetime curvature.

2.1.2 Deflection Angle

The total bending angle for a light ray passing at closest approach r_0 from a mass M follows from the geodesics of the acoustic metric:

$$\theta_{\text{total}} = \frac{4GM}{r_0\, c^2} \tag{7}$$

This can be decomposed into two contributions of equal magnitude: $2GM/(r_0 c^2)$ from transverse advection by the inflow (the light beam is carried sideways by the flowing medium) and $2GM/(r_0 c^2)$ from the time-delay asymmetry (the beam spends more coordinate time in the near-side flow than the far-side, creating a net deflection). Both contributions arise from the same inflow—they are aspects of a single physical mechanism, not two independent effects.

This matches the observed gravitational deflection confirmed since Eddington's 1919 solar eclipse measurements and subsequently verified to better than 0.01% precision by radio interferometry [10]. The Aether model reproduces the full general-relativistic bending angle from the acoustic metric of a flowing medium, without invoking spacetime curvature.

2.1.3 The Event Horizon

The inflow velocity $v_{\text{in}}(r) = \sqrt{2GM/r}$ increases as r decreases. At the Schwarzschild radius $r_s = 2GM/c^2$, the inflow reaches the speed of light:

$$v_{\text{in}}(r_s) = \sqrt{\frac{2GM}{2GM/c^2}} = c \tag{8}$$

At this radius, no outward-propagating wave can escape—not because spacetime geometry prevents it, but because the medium itself is flowing inward at the wave propagation speed. This is directly analogous to a sonic horizon in fluid dynamics, where a supersonic flow prevents upstream sound propagation.

Inside the Schwarzschild radius, the inflow exceeds c. The transient binding cycle still operates locally (pair formation and annihilation do not require outward propagation), but the restoring pressure pulses from pair annihilation can no longer propagate outward. The energy from annihilation is carried further inward by the flow, potentially leading to net Aether accumulation at the center.

2.1.4 Steady-State Conservation

The Aether inflow at escape velocity raises an immediate question: if Aether continuously flows inward toward mass, where does it go? Does the mass grow from accumulating Aether?

The answer lies in the transient binding cycle that the theory already describes. The inflow is not a separate process from gravity—it *is* gravity, viewed from the fluid-dynamics perspective:

1. Electron vortices draw Aether inward (creating the inflow).
2. The vortex core acts as a rapidly moving boundary whose circulation velocity exceeds the DCE threshold, producing transient particle–antiparticle pairs (consuming Aether from the local region).
3. The pair annihilates almost immediately, restoring the consumed Aether (returning it to the medium).
4. The restored Aether is available to flow inward again, completing the cycle.

The net Aether consumption is essentially zero—only the tiny asymmetry ϵ propagates outward as the gravitational wave. The inflow velocity $v_{\mathrm{in}}(r) = \sqrt{2GM/r}$ represents the bulk drift velocity of a steady-state circulation pattern, not a one-way accumulation. The kinetic energy of the inflow at radius r is $\frac{1}{2}\rho_A v_{\mathrm{in}}^2 = \rho_A GM/r$, exactly equal to the gravitational potential energy density—consistent with free-fall into a gravitational potential well with full energy conservation.

This picture connects to the vacuum catastrophe: the enormous energy of vacuum fluctuations (approximately 10^{120} times the cosmological constant) is the total cycling energy being invested and recovered at every point in space. The gravitational effect comes only from the tiny unrecycled asymmetry. The Aether inflow is the macroscopic manifestation of this microscopic recycling process.

This theory also predicts that gravitational waves propagate at the speed of light, since light speed represents the propagation speed of compression waves in Aether.

3. Summary of Theoretical Properties

3.1 Properties of Gravity

- An oscillating pressure wave (normal $\longrightarrow$ low $\longrightarrow$ normal) produced by the transient bind–unbind cycling of Aether at every atom, corresponding to virtual particle–antiparticle fluctuations in quantum field theory.
- Triggered by the electron vortex core acting as a rapidly moving boundary—the DCE mechanism operating below the permanence threshold.
- At macroscopic scales, the extraordinarily high cycling frequency makes this oscillation indistinguishable from a static attractive field.

- Propagates at the speed of light.
- Requires no net consumption of Aether and no ejection of particles. Energy is recycled locally; only the pressure wave propagates outward.
- The weakness of gravity relative to other forces reflects the fact that nearly all energy in each cycle is recycled, with only a tiny asymmetry (the difference between the low-pressure dip and the normal-pressure restoration) propagating as the wave.
- The gravitational constant G encodes the transient particle mass, cycling frequency, vortex velocity threshold, and transient/permanent ratio (DCE threshold), all constrained by the Heisenberg uncertainty principle ($\Delta E \cdot \Delta t \geq \hbar/2$) and the DCE mathematics. G is predicted to be derivable from the Coulomb constant k, the speed of light c, and the asymmetry fraction ϵ of the bind–unbind cycle—reducing all force constants to properties of the same Aether medium.
- The Aether inflow velocity at radius r is $v_{\text{in}}(r) = \sqrt{2GM/r}$, equal to the Newtonian escape velocity. At the Schwarzschild radius $r_s = 2GM/c^2$, the inflow reaches c, producing a sonic horizon (event horizon) beyond which no outward-propagating wave can escape. The inflow is a steady-state circulation: Aether flows in, participates in transient binding, and is returned upon pair annihilation. In the two-fluid model (Paper 6 [13]), the gravitational inflow is the center-of-mass velocity of both the superfluid and normal components flowing inward together; the bind–unbind cycle operates on the superfluid condensate, which constitutes ~100% of the Aether near matter where $T \ll T_c$.
- Self-limiting feedback, governed by the uncertainty principle and the DCE requirement for sustained vortex motion, prevents runaway cascading of transient binding events.
- **Gravitomagnetism.** A rotating mass drags bound Aether into co-rotation, creating angular flow structure $\mathbf{B}_g = -(1/c^2)\,\mathbf{v} \times \mathbf{g}$—the gravitational flow angle, analogous to the electromagnetic flow angle, governed by the same v^2/c^2 ratio. Confirmed by GP-B (frame-dragging precession 39.3 vs. measured 37.2 ± 7.2 mas yr^{-1}).

4. Experimental Support and Connections to Established Physics

The following experimentally confirmed phenomena support or align with the aspects of the theory presented in this paper:

1. **Cherenkov radiation.** Particles exceeding the local speed of light in a material medium produce light—consistent with this theory's prediction that bound matter modifies the Aether's wave propagation properties (reducing the effective wave speed to c/n), creating conditions for a compression shockwave.

2. **Gravitational wave speed.** LIGO's detection of gravitational waves [4], and the near-simultaneous arrival of gravitational and electromagnetic signals from the neutron-star merger GW170817 [11], confirmed they propagate at the speed of light to within $|\Delta c/c| < 10^{-15}$, consistent with this theory's prediction that gravitational waves are compression waves in the same medium that carries light.

3. **Gravitational lensing bending angle.** The observed deflection of light by massive objects is $4GM/(r_0c^2)$, confirmed since Eddington's 1919 solar eclipse [2] and verified to better than 0.01% by radio tracking [10]. This theory derives this exact result from the acoustic metric of the Aether inflow at escape velocity $v_{\text{in}} = \sqrt{2GM/r}$, which takes the Painlevé–Gullstrand form and is exactly equivalent to the Schwarzschild geometry of general relativity.

4. **Gravitomagnetic frame dragging (GP-B, 2011; LAGEOS, 2004).** The v^2/c^2 universality applied to gravity predicts a frame-dragging precession of 39.3 mas yr^{-1} for the GP-B orbit (Eq. 5), with no adjustable parameters. GP-B measured -37.2 ± 7.2 mas yr^{-1}, consistent within 0.3σ. LAGEOS confirmed frame dragging independently to $\sim 10\%$ precision [3, 9].

5. **Null results for static electromagnetic–gravitational coupling.** Hathaway and Williams [7] found no weight change for conducting objects at potentials up to 200 kV (null to within 2 g), and Tajmar, Kössling, and Neunzig [8] tested capacitors, solenoids, and tunneling-current devices on high-resolution balances in vacuum, finding no anomalous forces down to ~3 μN. The Aether model predicts exactly this null result: static electric fields are the time-averaged coherent axial-oscillation wave field of the bound core ring (Paper 4 [14]), a net-zero-flow oscillation that does not involve the transient bind–unbind cycling that generates gravity. Only time-varying disruptions of the cycling process can modify gravitational output.

5. Open Questions and Testable Predictions

The following areas relevant to this paper remain underdeveloped and represent opportunities for further refinement or falsification:

1. **Deriving G from fundamental constants.** If transient binding corresponds to virtual particle fluctuations governed by the Heisenberg uncertainty principle, and the transient/permanent ratio is governed by the DCE threshold, it may be possible to derive the gravitational constant from $\hbar$, c, the DCE threshold velocity, and the properties of Aether. A successful derivation would be strong evidence for this theory and would represent a unification of gravity and quantum mechanics.

2. **The vacuum catastrophe ratio.** This theory predicts that the ratio of total vacuum fluctuation energy to the observed cosmological constant should equal the ratio of total cycling energy

to the asymmetric residual that propagates as gravity. If this ratio can be calculated from the theory's parameters, it could resolve the vacuum catastrophe—one of the largest unsolved problems in physics.

3. **Gravitational wave frequency.** If gravity is an oscillating wave rather than a static field, it has a characteristic frequency determined by the cycling rate. The uncertainty principle constrains this frequency. If it is finite (however high), it is in principle detectable. Identifying this frequency would be a strong confirmation of the theory.

4. **Variation of G.** If G depends on local Aether conditions (density, temperature), precision measurements of G in different environments could reveal deviations from constancy. Such deviations, if found, would support this theory over standard models where G is a universal constant.

5. **Steady-state Aether inflow and interior solutions.** The gravitational inflow velocity $v_{\text{in}}(r) = \sqrt{2GM/r}$ reaches c at the Schwarzschild radius, producing a sonic horizon from which outward-propagating waves cannot escape. Inside this radius, the transient binding cycle still operates but the restoring pressure pulses are carried further inward rather than propagating outward, potentially leading to net Aether accumulation at the center. The conservation analysis (steady-state recycling at the macroscopic level) should be extended to include the gravitational wave energy loss rate ϵ and its long-term effect on mass evolution.

6. **Self-consistent strong-field solution.** The weak-field inflow velocity $v_{\text{in}}(r) = \sqrt{2GM/r}$ is a first-order approximation valid when $2GM/(rc^2) \ll 1$. Near a black hole, where v_{in} approaches c, the full solution requires solving the inflow velocity profile, the acoustic metric, and the bind–unbind cycling rate self-consistently through the continuity and momentum equations (with the Aether's logarithmic equation of state). The Bernoulli analysis guarantees $\Delta\rho_A/\rho_A = 0$ in the weak field; the strong-field regime must determine whether this holds or whether relativistic corrections introduce density perturbations. This self-consistent solution would determine: (a) the exact location of the sonic horizon, (b) the precise photon sphere radius, (c) whether the Aether acoustic metric reproduces the Schwarzschild metric in the strong-field regime, and (d) the exact Hawking temperature coefficient.

7. **Gravitational wave merger waveforms.** LIGO and Virgo [4] have detected dozens of binary black hole mergers whose gravitational waveforms match general-relativistic numerical simulations with extraordinary precision, including the inspiral chirp, the merger peak, and the ringdown quasi-normal modes. The Aether model predicts gravitational waves propagate at c (confirmed) and that they arise from compression waves in the medium. Reproducing the detailed merger waveforms—particularly the nonlinear merger and ringdown phases—from Aether dynamics would be a stringent test. This likely requires numerical simulation of colliding Aether inflow patterns.

8. **Electromagnetic control of the asymmetry fraction ϵ.** The gravitational constant $G \propto \epsilon\, \rho_A\, \psi^2$ depends on the asymmetry fraction of the bind–unbind cycle. Can ϵ be modified by external electromagnetic fields? The null results for static fields [7, 8] constrain the coupling, but time-varying fields that resonantly drive the binding cycle have not been tested. Determining whether ϵ is a fixed property of the medium or a dynamically controllable parameter would have profound implications for the relationship between gravity and electromagnetism.

9. **Cooper pair suppression of the gravitational cycling rate.** In a superconductor, electrons form Cooper pairs whose opposite-spin vortices cancel all net Aether flow. Does this cancellation extend to the transient bind–unbind cycle? If Cooper pairing suppresses the cycling rate or modifies ϵ within the condensate, a superconductor should exhibit a measurable difference in gravitational coupling compared to the same material above its critical temperature. Precision measurements of the gravitational attraction between a test mass and a sample cycled above and below T_c could test this prediction.

6. Proposed Experiments

The following experiments target specific predictions relevant to this paper:

1. **Weight measurement above a driven superconductor with AC fields.** The theory predicts that static electromagnetic fields cannot modify gravity (confirmed by null results [7, 8]), but that *time-varying* fields applied to a superconductor could coherently drive the transient bind–unbind cycle, amplifying the asymmetry fraction ϵ and producing a measurable weight change in a test mass above the device. The proposed setup consists of a large-diameter YBCO or niobium superconductor subjected to alternating magnetic fields at frequencies from kHz to GHz, with a precision balance measuring test mass weight changes under vacuum. The theory predicts: (a) no effect from static fields (already confirmed), (b) a frequency-dependent onset when the driving field couples to the bind–unbind cycling rate, and (c) the effect should vanish above the superconducting critical temperature, since Cooper pair coherence is required to make the cycling collective rather than random.

2. **Refractive-index gradient as a gravitational potential.** In the Aether model, a material with refractive index n increases the effective Aether density by n^2 within its volume, reducing the local wave speed to c/n. (This is distinct from gravitational fields, where Bernoulli gives $\Delta\rho_A = 0$.) A metamaterial with an engineered refractive-index gradient therefore creates a gradient in local c, which is physically identical to a gravitational potential gradient. The proposed experiment uses a metamaterial slab with a monotonic refractive-index gradient from $n = 1$ to $n \approx 2$ over a distance of $\sim$10 cm, and measures clock rates (via precision atomic spectroscopy or optical frequency comparison) at different positions within the gradient.

The theory predicts a fractional frequency shift of $\Delta f/f = (n_2 - n_1)/n_2$ between the two positions—since clock rates scale with the local wave speed c/n, the fractional shift is simply the ratio of local propagation speeds.

3. **Asymmetric waveform gravity generation.** The Aether model predicts gravity arises from an asymmetric pressure wave: the bind event produces a low-pressure pulse, and the unbind event partially restores pressure, but with a net deficit (the asymmetry fraction ϵ). An artificial gravity source could in principle be constructed by driving an asymmetric electromagnetic waveform—fast compression followed by slow rarefaction—through a medium capable of coupling the electromagnetic oscillation to local Aether density changes. A superconductor is the natural candidate, since its Cooper pair condensate couples coherently to the medium. The proposed experiment subjects a superconductor to a sawtooth-wave magnetic field (sharp rise, slow fall) and measures whether the asymmetric driving produces a net force on nearby test masses, compared to a symmetric sinusoidal drive at the same frequency and amplitude (which the theory predicts should produce no net effect).

4. **Casimir cavity gravitational test.** The dynamical Casimir effect has been confirmed in superconducting circuits [1], demonstrating that accelerating boundaries convert vacuum fluctuations into real particles. This is the microscopic mechanism by which the theory proposes mass forms from Aether. The proposed experiment measures whether a Casimir cavity undergoing rapid, asymmetric mechanical oscillation produces a detectable gravitational signal—either a static weight change or an oscillating force on a nearby test mass. The theory predicts that symmetric oscillation produces no net gravitational output (equal binding and unbinding), while asymmetric oscillation (where the boundary spends more time accelerating in one direction) should produce a net gravitational impulse proportional to the asymmetry and the DCE photon production rate.

5. **Superfluid helium analog of gravitational time dilation.** The 2024 observation of Kerr black hole geometry in a superfluid ^{4}He vortex [6] demonstrated that quantum superfluids reproduce gravitational spacetime structure. The next step is to measure the analog of *gravitational time dilation*: whether wave propagation through the radial inflow of a draining vortex exhibits time dilation consistent with the Painlevé–Gullstrand acoustic metric, where a static observer in the inflow experiences $\delta t/t = v_{\mathrm{in}}^2/(2c_s^2)$. Existing superfluid vortex facilities could perform this measurement by comparing wave arrival times at different radial distances from a stabilized giant vortex. Agreement would confirm that gravitational time dilation arises from the inflow velocity field of a superfluid medium.

6. **Frequency dependence of the gravitational constant.** If gravity arises from an oscillating bind–unbind cycle rather than a static field, measurements of G at different timescales could reveal the cycling frequency. Torsion balance measurements of G (which operate at periods of minutes) should agree with atomic interferometry measurements (which operate at millisecond

timescales). However, at frequencies approaching the bind–unbind cycling rate, the effective G should begin to decrease as the measurement period becomes shorter than the cycling period. Comparing precision measurements of G across the widest possible range of measurement timescales—from torsion balances to atom interferometers to high-frequency resonators—could reveal the characteristic frequency of the gravitational cycle.

For all proposed experiments, see Paper 1: General Overview [12].

7. Mathematical Summary

This section collects the key equations relevant to the topics covered in this paper. *For the complete mathematical summary, see Paper 1: General Overview [12].*

7.1 Fundamental Aether Parameters

The theory posits four microscopic parameters from which all macroscopic phenomena derive:

Symbol	Name	Constrained range	Primary constraint
ρ_A	Ambient density	10^{11}–10^{12} kg/m^3	Electron vortex energy
K_A	Bulk modulus	10^{28}–10^{29} J/m^3	$K_A = \rho_A c^2$
m_A	Aether quantum mass	500–850 MeV/c^2	Flux tube width / glueball mass
ξ	Healing length	0.17–0.28 fm	Bridge diameter (lattice QCD)

These four are not independent. Two defining relations connect them:

$$c = \sqrt{K_A/\rho_A} \tag{9}$$

$$\xi = \frac{\hbar}{m_A\, c\, \sqrt{2}} \tag{10}$$

leaving two free Aether parameters (e.g., ρ_A and m_A) from which all others follow. (A third parameter, ϵ, enters through gravity; see below.)

7.2 Equation of State

The Aether obeys a logarithmic equation of state, the unique form that produces a density-independent bulk modulus (Section 4):

$$P = K_A \ln\!\left(\frac{\rho}{\rho_0}\right) + P_0 \tag{11}$$

Key consequence: the speed of sound $c_s = \sqrt{K_A/\rho}$ varies with density. At ambient density, $c_s = c$. Note: in gravitational fields, Bernoulli's equation for free-fall inflow gives $\Delta\rho/\rho_A = 0$ exactly, so c

does not vary near gravitating masses; the density dependence is relevant for material media and for the two-fluid cosmological structure. This equation of state guarantees that K_A is constant under compression, which is experimentally confirmed by:

- Velocity time dilation measurements spanning $10^{-6}\,c$ to $0.9994\,c$ (precision $\sim 10^{-8}$).
- The Fresnel drag coefficient $1 - 1/n^2$, exact across materials with density ratios up to ~6:1.

7.3 Gravity

Gravity arises from the residual asymmetry of transient bind–unbind cycling.

Inflow velocity.

$$v_{\text{in}}(r) = \sqrt{\frac{2GM}{r}} \tag{12}$$

Density perturbation from gravity. Bernoulli's equation for free-fall inflow gives $\Delta\rho/\rho_A = 0$ exactly: the kinetic energy of the inflow balances the gravitational potential, leaving no energy for compression.

Gravitational time dilation (from inflow velocity).

$$\frac{\Delta t}{t} = \frac{v_{\text{in}}^2}{2c^2} = \frac{GM}{r\,c^2} \tag{13}$$

A static observer in the inflowing Aether experiences velocity time dilation (Painlevé–Gullstrand result).

Event horizon (sonic horizon).

$$r_s = \frac{2GM}{c^2}, \qquad v_{\text{in}}(r_s) = c \tag{14}$$

Gravitational lensing. The full bending angle from the acoustic metric of the inflowing Aether (transverse advection plus time-delay asymmetry):

$$\theta = \frac{4GM}{r_0\,c^2} \tag{15}$$

Force constant hierarchy.

$$G = \epsilon \, \frac{k \text{ (geometric and mass factors)}}{c^2} \tag{16}$$

where $\epsilon \ll 1$ is the fractional asymmetry of the bind–unbind cycle, encoding the electromagnetic-to-gravitational force ratio between two electrons, $ke^2/(Gm_e^2) \approx 4.17 \times 10^{42}$.

Hawking temperature. From the inflow velocity gradient at the sonic horizon:

$$T_H = \frac{\hbar\, c^3}{8\pi\, G\, M\, k_B} \tag{17}$$

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